

MH1100 Calculus I

Handout 5 - Continuity

Synopsis

The definition of and topics related to the important concept of **Continuity**.

- The definition and first examples
- Using elementary operations to build new continuous functions from other continuous functions
- Important classes of continuous functions: Polynomials, Power functions, Rational functions, Algebraic functions, Trigonometric functions, Exponential functions, Logarithmic functions
- Compositions
- The Intermediate Value Theorem

Continuity

We noticed previously that the limit of a function as x approaches a can often be found simply by calculating the value of the function at a .

We know that

Direct Substitution Property

If f is a polynomial or a rational function and a is in the domain of f , then

$$\lim_{x \rightarrow a} f(x) = f(a).$$

For example, if

$$f(x) = \frac{x+1}{x^2-2},$$

then

$$\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} \frac{x+1}{x^2-2} = f(1) = -2.$$

Continuity

It is true that for many functions $f(x)$ we can obtain the limit $\lim_{x \rightarrow a} f(x)$ just by evaluating the function at the point we are taking the limit: $f(a)$.

These are actually very special types of functions: the **continuous** functions.

Definition

Let f be a function and let $a \in \mathbb{R}$. The function f is said to be **continuous at the point a** if the following three things are true:

- $f(a)$ is defined (that is, a is in the domain of f). *$a \in \text{Dom}(f)$*
- The limit $\lim_{x \rightarrow a} f(x)$ exists.
- $\lim_{x \rightarrow a} f(x) = f(a)$.

Continuity

Definition

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- $\lim_{x \rightarrow a} f(x) = f(a)$.

Comments:

- Often we **summarize** these three conditions by a **single equation**

$$\lim_{x \rightarrow a} f(x) = f(a).$$

- This equation is sometimes called the **direct substitution property**.
- The intuitive idea of a continuous function is that *the graph can be drawn without lifting the pen off the paper*. Or - a small change in x away from $x = a$ only leads to a small change in $f(x)$.

Continuity

Example A

Use the definition of continuity and standard properties of limits to show that the function $f(x) = x^2 + \sqrt{5-x}$ is continuous at the point $x = 4$.

Solution $\text{Dom}(f) = (-\infty, 5]$

$\therefore 4 \in \text{Dom}(f)$

$$\begin{aligned}\lim_{x \rightarrow 4} f(x) &= \lim_{x \rightarrow 4} (x^2 + \sqrt{5-x}) = \lim_{x \rightarrow 4} x^2 + \lim_{x \rightarrow 4} \sqrt{5-x} \\ &= \left(\lim_{x \rightarrow 4} x\right)^2 + \sqrt{\lim_{x \rightarrow 4} (5-x)} = 4^2 + \sqrt{1} = 17 \\ &= f(4).\end{aligned}$$

} using limit laws

We can conclude that f is continuous at $x=4$, because:

1. f is defined at $x=4$
2. The limit of f at $x=4$ exists.
3. The limit equals the value of f at $x=4$.

Discontinuity

If f is defined near a (in other words, f is defined on an open interval containing a , except perhaps at a), we say that f is **discontinuous at a** (or f has a **discontinuity** at a) if f is not continuous at a .

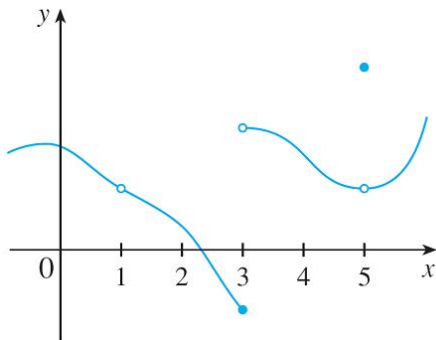
Comments: if f is NOT continuous at a , then

- $f(a)$ is NOT defined (that is, a is NOT in the domain of f).
- Or the limit $\lim_{x \rightarrow a} f(x)$ does NOT exist.
- Or $\lim_{x \rightarrow a} f(x) \neq f(a)$.

Continuity

Exercise

Figure below shows the graph of a function f . At which numbers is f discontinuous? Why?



$x=1$
 $x=3$
 $x=5$ } discontinuous at these points.

Function is continuous on all other points

→ for ending point, we consider continuity from one side, so they are continuous.

Example 2

Where are each of the following functions discontinuous?

$$(a) f(x) = \frac{x^2 - x - 2}{x - 2}$$

$$(b) f(x) = \begin{cases} \frac{1}{x^2}, & \text{if } x \neq 0 \\ 1, & \text{if } x = 0 \end{cases}$$

$$(c) f(x) = \begin{cases} \frac{x^2 - x - 2}{x - 2}, & \text{if } x \neq 2 \\ 1, & \text{if } x = 2 \end{cases}$$

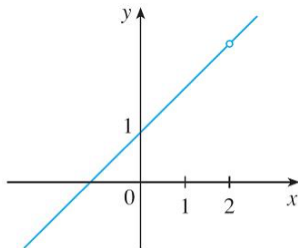
$$(d) f(x) = \lfloor x \rfloor = \text{The largest integer that is less than or equal to } x.$$

$f(x) = \lfloor x \rfloor$ is the **greatest integer function**. Sometimes it is also called the **floor function**. For instance, $\lfloor 4 \rfloor = 4$, $\lfloor 4.8 \rfloor = 4$, $\lfloor \pi \rfloor = 3$, $\lfloor \sqrt{2} \rfloor = 1$, and $\lfloor -0.5 \rfloor = -1$.

Example 2–Solution

Solution

(a) Notice that $f(2)$ is not defined, so f is discontinuous at 2. Later we will see why f is continuous at all other numbers.



$$(a) f(x) = \frac{x^2 - x - 2}{x - 2} = \frac{(x-2)(x+1)}{x-2} = x+1$$

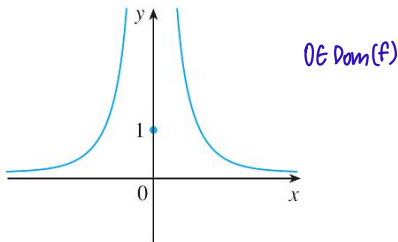
The kind of discontinuity is called **removable** because we could remove the discontinuity **by redefining f** at just the single number 2.

Example 2–Solution (Cont'd)

(b) Here $f(0) = 1$ is defined, but

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$$

does not exist. So f is discontinuous at $x = 0$.



$$(b) f(x) = \begin{cases} \frac{1}{x^2} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$$

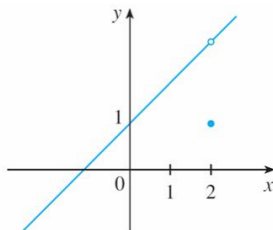
The discontinuity in part (b) is called an **infinite discontinuity**.

Example 2–Solution (Cont'd)

(c) Here $f(2) = 1$ is defined and

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x - 2} = \lim_{x \rightarrow 2} \frac{(x - 2)(x + 1)}{x - 2} = \lim_{x \rightarrow 2} (x + 1) = 3$$

exists. But $\lim_{x \rightarrow 2} f(x) = 3 \neq 1 = f(2)$. So f is discontinuous at $x = 2$.



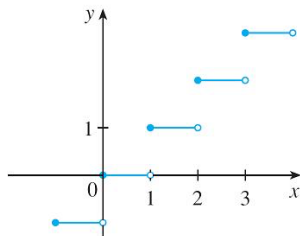
$f(2) = 1$
 $\text{Dom}(f) = \mathbb{R}$

$$(c) f(x) = \begin{cases} \frac{x^2 - x - 2}{x - 2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$$

The discontinuity in part (c) is called a **removable discontinuity**.

Example 2–Solution (Cont'd)

(d) The greatest integer function $f(x) = \lfloor x \rfloor$ has discontinuities at all of the integers because $\lim_{x \rightarrow n} \lfloor x \rfloor$ does not exist if n is an integer.



$$\lim_{x \rightarrow n^+} \lfloor x \rfloor = n$$

$$\lim_{x \rightarrow n^-} \lfloor x \rfloor = n-1$$

(d) $f(x) = \lfloor x \rfloor$

The discontinuities in part (d) is called **jump discontinuity** because the function **jump** from one value to another.

One-sided continuity

Just like one-sided limits, we can also define one-sided continuities.

Definition

A function f is continuous from the right at a number a if

$$\lim_{x \rightarrow a^+} f(x) = f(a)$$

and f is continuous from the left at a number a if

$$\lim_{x \rightarrow a^-} f(x) = f(a)$$

One-sided continuity

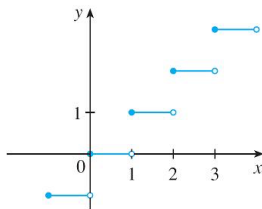
Example 3

At each integer n , the greatest integer function $f(x) = \lfloor x \rfloor$ is continuous from the right because

$$\lim_{x \rightarrow n^+} f(x) = \lim_{x \rightarrow n^+} \lfloor x \rfloor = n = f(n)$$

but discontinuous from the left because

$$\lim_{x \rightarrow n^-} f(x) = \lim_{x \rightarrow n^-} \lfloor x \rfloor = n - 1 \neq n = f(n)$$



Continuity on an interval

So far we have only been discussing continuity at a single point.

Functions which are continuous at every point of a whole interval have some really nice properties. To discuss these properties precisely it is useful to introduce another definition.

Definition: Continuous on an interval

A function f is **continuous on an interval** if it is **continuous at every number in the interval**.

(If f is defined only on one side of an endpoint of the interval, we understand *continuous* at the endpoint to mean *continuous from the right* or *continuous from the left*.)  $f(x) = \sqrt{x} \quad x \in [0, \infty)$

 Note that by **interval** we are including **all** of the following possibilities: (a, b) , $(a, b]$, $[a, b)$, $[a, b]$, $(-\infty, b)$, $(-\infty, b]$, (a, ∞) , $[a, \infty)$, and $(-\infty, \infty)$.
no need to discuss the end points

Continuity on an interval

Exercise

$$1-x^2 \geq 0 \rightarrow x^2 \leq 1$$

Show that the function $f(x) = 2 - \sqrt{1-x^2}$ is continuous on the interval $[-1, 1]$.

$$\text{Dom}(f) = [-1, 1]$$

If $-1 < a < 1$, then using the limit laws, we have

$$\begin{aligned} \text{Case 1: } \lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} (2 - \sqrt{1-x^2}) = 2 - \lim_{x \rightarrow a} \sqrt{1-x^2} \\ &= 2 - \sqrt{\lim_{x \rightarrow a} (1-x^2)} = 2 - \sqrt{1-a^2} \\ &= f(a) \end{aligned}$$

Thus, f is continuous at a if $-1 < a < 1$. Similar calculations show that

$$\text{Case 2: } \lim_{x \rightarrow -1^+} f(x) = 2 = f(-1) \quad \text{and} \quad \text{Case 3: } \lim_{x \rightarrow 1^-} f(x) = 2 = f(1)$$

So f is continuous from the right at -1 and continuous from left at 1 .

Therefore, f is continuous on $[-1, 1]$.

Continuity

Instead of always using definitions to verify the continuity of a function, it is often convenient to use the next theorem, which shows how to build up complicated continuous functions from simple ones by doing additions, multiplications, and divisions.

Theorem 4

If f and g are continuous at a and if c is a constant, then the following functions are also continuous at a :

1. $f + g$
2. $f - g$
3. cf
4. fg
5. $\frac{f}{g}$ if $g(a) \neq 0$

Use limit laws and continuity of f and g to prove the continuity of these 5 functions.

Comments:

- Each of the five parts of this theorem follows from the corresponding Limit Law in Section 1.6.
- If f and g are continuous on an interval, then so are the functions $f + g$, $f - g$, cf , fg , and f/g (if g is never 0).

Continuity

In Section 1.6 we have discussed the Direct Substitution Property of polynomials and rational functions.

Direct Substitution Property

If f is a polynomial or a rational function and a is in the domain of f , then

$$\lim_{x \rightarrow a} f(x) = f(a).$$

This suggests the following theorem

Theorem 5

- (a) Any polynomial is **continuous everywhere**; that is, it is continuous on $\mathbb{R} = (-\infty, \infty)$
- (b) Any rational function is continuous wherever it is defined; that is, it is continuous on its domain.

Continuity

Proof of Part (a) in Theorem 5

A polynomial is a function of the form

$$P(x) = c_n x^n + c_{n-1} x^{n-1} + \cdots + c_1 x + c_0 = \sum_{i=0}^n c_i x^i$$

where c_i ($i = 0, 1, \dots, n$) are constants.

Let $a \in \mathbb{R}$. The domain of any polynomial function is $\mathbb{R}$, so it is obvious that a will lie in the domain of the given $P(x)$.

Next consider the limit

$$\begin{aligned} \lim_{x \rightarrow a} P(x) &= \lim_{x \rightarrow a} \left(\sum_{i=0}^n c_i x^i \right) \stackrel{\text{sum law}}{=} \sum_{i=0}^n \lim_{x \rightarrow a} c_i x^i \\ &= \sum_{i=0}^n c_i \lim_{x \rightarrow a} x^i \stackrel{\text{power law}}{=} \sum_{i=0}^n c_i a^i = P(a) \end{aligned}$$

$\nwarrow$
constant multiple law

Thus the limit exists, and equals $P(a)$, as required by the definition.

Continuity

Knowledge of which functions are continuous enables us to evaluate some limits very quickly.

Exercise

Find $\lim_{x \rightarrow -2} \frac{x^3 + 3x^2 - 1}{5 - 3x}$.

The function

$$f(x) = \frac{x^3 + 3x^2 - 1}{5 - 3x}$$

is rational, so by Theorem 5 it is continuous on its domain, which is $\{x \in \mathbb{R} \mid x \neq \frac{5}{3}\}$

Therefore

$$\lim_{x \rightarrow -2} \frac{x^3 + 3x^2 - 1}{5 - 3x} = \lim_{x \rightarrow -2} f(x) = f(-2) = \frac{(-2)^3 + 3(-2)^2 - 1}{5 - 3(-2)} = \frac{3}{11}$$

Continuity

if $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} f(x) = M$

Prove $L = M$.

Assume $L \neq M$ (or $L > M$)

For $\varepsilon = \frac{L-M}{2}$, $\exists \delta > 0$

if $x \in (a - \delta, a + \delta) \setminus \{a\}$, then $|f(x) - M| < \frac{L-M}{2}$

$$\downarrow$$
$$M - \frac{L-M}{2} < f(x) < \frac{L-M}{2} + M$$

$\downarrow$

Will eventually get a contradiction

Youtube:

Proof: Suppose $L \neq M$

$\varepsilon > 0$, $\delta_1, \delta_2 > 0$, s.t. $\forall x$

$$|x - a| < \delta_1 \Rightarrow |f(x) - L| < \frac{|L-M|}{2} \quad \varepsilon$$

$$|x - a| < \delta_2 \Rightarrow |f(x) - M| < \frac{|L-M|}{2} \quad \varepsilon$$

Let $\delta = \min\{\delta_1, \delta_2\}$

$\varepsilon > 0$, $\delta > 0$ s.t. $\forall x$

$$|x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon \text{ and } |f(x) - M| < \varepsilon$$

$$\begin{aligned} |L - M| &= |L - f(x) + f(x) - M| \\ &\leq |L - f(x)| + |f(x) - M| \\ &= |f(x) - L| + |f(x) - M| \\ &< \frac{|L - M|}{2} + \frac{|L - M|}{2} \\ &< |L - M| \end{aligned}$$

This is a contradiction

Continuity

It turns out that most of the familiar functions are continuous at every number in their domains. How about sine and cosine functions?

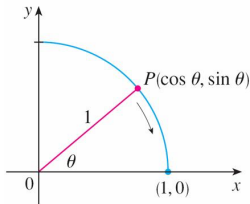
We know from the definitions of $\sin \theta$ and $\cos \theta$ that the coordinates of the point P in the figure below are $(\cos \theta, \sin \theta)$. As $\theta \rightarrow 0$, we see that P approaches the point $(1, 0)$ and thus

$$\lim_{\theta \rightarrow 0} \cos \theta = 1$$

$$\lim_{\theta \rightarrow 0} \sin \theta = 0$$

$$\lim_{\theta \rightarrow 0} \cos \theta = \cos 0$$

$$\lim_{\theta \rightarrow 0} \sin \theta = \sin 0$$



Since $\cos 0 = 1$ and $\sin 0 = 0$, the above equations assert that the cosine and sine functions are continuous at 0. The addition formulas for cosine and sine can then be used to deduce that these functions are continuous everywhere.

Continuity

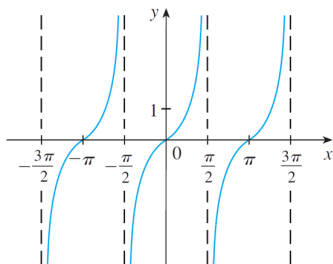
It follows from part 5 of Theorem 4 that

$$\tan x = \frac{\sin x}{\cos x}$$

continuous everywhere (pointing to $\sin x$)
continuous everywhere (pointing to $\cos x$)

is continuous except where $\cos x = 0$. This happens when x is an odd integer multiple of $\pi/2$, so $y = \tan x$ has infinite discontinuities when $x = \pm\pi/2, \pm3\pi/2, \pm5\pi/2$, and so on.

cos x will be 0.



Continuity

The essential functions summarized in §1.2 are continuous in their domains.

Theorem 7

The following functions are continuous at every number in their domains:

- Polynomials $P(x) = \sum_{i=0}^n c_i x^i$
- Power functions $f(x) = x^a$
- Rational functions $f(x) = P(x)/Q(x)$
- Algebraic functions
- Trigonometric functions $\sin x, \cos x, \tan x, \csc x, \sec x, \text{ and } \cot x$
- Exponential functions $f(x) = b^x \ (b > 0)$
- Logarithmic functions $f(x) = \log_b x \ (b > 0)$

The general principle: Any function constructed from continuous functions by doing additions, multiplications, and divisions, is itself going to be continuous at every point of its domain.

Continuity

Another way of combining continuous functions f and g to get a new continuous function is to form the composite function $f \circ g$. This fact is a consequence of the following theorem.

Theorem 8

If f is continuous at b and $\lim_{x \rightarrow a} g(x) = b$, then $\lim_{x \rightarrow a} f(g(x)) = f(b)$. In other words, ①

$$\lim_{x \rightarrow a} f(g(x)) = f(\lim_{x \rightarrow a} g(x))$$

This theorem says that a limit symbol $\lim_{x \rightarrow a}$ can be moved through a function symbol f if the function is continuous and the limit exists. In other words, the order of these two symbols can be reversed.

Intuitively, Theorem 8 is reasonable because if x is close to a , then $g(x)$ is close to b , and since f is continuous at b , if $g(x)$ is close to b , then $f(g(x))$ is close to $f(b)$.

Continuity

Exercise

Prove the Root Law.

Root Law

Let g be a function and let $a \in \mathbb{R}$. Suppose that n is a positive integer. If the limit $\lim_{x \rightarrow a} g(x)$ exists, then $\lim_{x \rightarrow a} \sqrt[n]{g(x)}$ exists and

$$\lim_{x \rightarrow a} \sqrt[n]{g(x)} = \sqrt[n]{\lim_{x \rightarrow a} g(x)}.$$

If n is even, we assume that $\lim_{x \rightarrow a} g(x) > 0$.

Proof: Let $f(x) = \sqrt[n]{x}$, with n being a positive number. Then

$$f(g(x)) = \sqrt[n]{g(x)}$$

and

$$f\left[\lim_{x \rightarrow a} g(x)\right] = \sqrt[n]{\lim_{x \rightarrow a} g(x)}$$

If we put these expressions into Theorem 5, we get

$$\lim_{x \rightarrow a} \sqrt[n]{g(x)} = \sqrt[n]{\lim_{x \rightarrow a} g(x)}$$

Continuity

A continuous function f of a continuous function g is a continuous function $f \circ g$.

Theorem 9

If g is continuous at a and f is continuous at $g(a)$, then the composite function $f \circ g$ given by $(f \circ g)(x) = f(g(x))$ is continuous at a .

Proof. Since g is continuous at a , we have

A continuous function of a continuous function is continuous.

$$\lim_{x \rightarrow a} g(x) = g(a)$$

Since f is continuous at $b = g(a)$, we can apply Theorem 8 to obtain

$$\lim_{x \rightarrow a} f(g(x)) = f(\lim_{x \rightarrow a} g(x)) = f(g(a))$$

which is precisely the statement that the function $h(x) = f(g(x))$ is continuous at a ; that is, $f \circ g$ is continuous at a .

Example 8

Where is the following function continuous?

$$F(x) = \frac{1}{\sqrt{x^2 + 7} - 4}$$

$\rightarrow$ we already know it's continuous

$$\lim_{x \rightarrow 1} F(x) = F(1) = \frac{1}{2\sqrt{2} - 4}$$

Solution. Note that F can be broken up as the composition of four continuous functions:

$$F = f \circ g \circ h \circ k \quad \text{or} \quad F(x) = f(g(h(k(x))))$$

where

$$f(x) = \frac{1}{x} \quad g(x) = x - 4 \quad h(x) = \sqrt{x} \quad k(x) = x^2 + 7$$

We know that each of these functions is continuous on its domain, so by Theorem 9, F is continuous on its domain, which is

$$\{x \in \mathbb{R} \mid \sqrt{x^2 + 7} \neq 4\} = \{x \in \mathbb{R} \mid x \neq \pm 3\} = (-\infty, -3) \cup (-3, 3) \cup (3, \infty).$$

Continuity

More generally, a function (such as a rational function) may have a limit at a point where it is not defined. If $f(a)$ is not defined, but $\lim_{x \rightarrow a} f(x) = L$ exists, we can define a new function $F(x)$ by the rule

$$F(x) = \begin{cases} f(x), & \text{if } x \text{ is in the domain of } f \\ L, & \text{if } x = a \end{cases}$$

The function F is continuous at $x = a$. It is called the **continuous extension of f to $x = a$** . For rational functions f , continuous extensions are often found by cancelling common factors in the numerator and denominator.

Exercise

$$f(x) = \frac{(x-3)(x+2)}{(x-3)(x+3)} = \frac{x+2}{x+3} \quad (x \neq 3)$$

Show that

$$f(x) = \frac{x^2 - x - 6}{x^2 - 9}, \quad x \neq 3$$

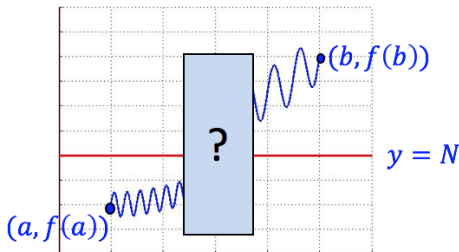
$$\lim_{x \rightarrow 3} f(x) = \frac{5}{6}$$

has a continuous extension to $x = 3$, and find the extension.

We cannot do continuous extension at $x = -3 \Rightarrow$ it has a infinite limit.

Continuity

Consider a function $f(x)$ with domain $[a, b]$ whose graph is shown below. I have concealed part of the graph under a box.



Question.

If I tell you that at no point does the graph of $f(x)$ intersect the line $y = N$ then what can you conclude?

Answer.

The function $f(x)$ cannot be continuous on the whole interval $[a, b]$.

Intermediate Value Theorem

The **observation** we just made is actually an important theorem about an important property of continuous functions:

*squeeze theorem $\rightarrow$ for limits
mean value theorem $\rightarrow$ for differential functions
 $\rightarrow$ can derive L'Hopital's rule*

$\rightarrow$ for continuous functions

The Intermediate Value Theorem *IVT*

Let f be a function, and let a , b , and N be real numbers with $a < b$.

Assume that:

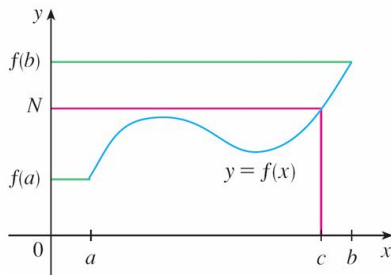
- f is **continuous** on the closed interval $[a, b]$
- $f(a) \neq f(b)$
- N lies between $f(a)$ and $f(b)$. $\begin{cases} f(a) < N < f(b) \\ f(b) < N < f(a) \end{cases}$

Then there exists a number c such that

- $a < c < b$ $c \in [a, b]$
- $f(c) = N$

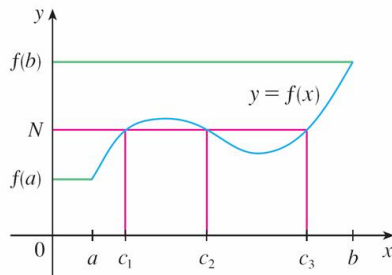
Intermediate Value Theorem

The Intermediate Value Theorem states that a continuous function takes on every intermediate value between the function values $f(a)$ and $f(b)$.



$$f(c) = N$$

(a)

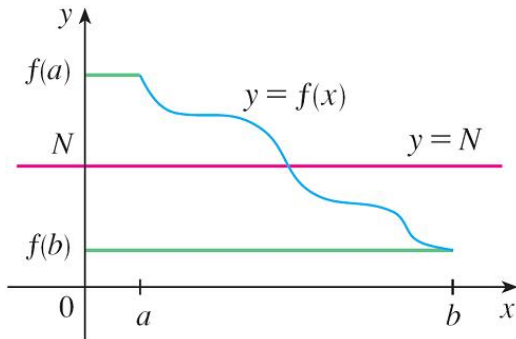


(b)

Note that the value N can be taken on once [as in part (a)] or more than once [as in part (b)].

Intermediate Value Theorem

If we think of a continuous function as a function whose graph has no hole or break, then it is easy to believe that the Intermediate Value Theorem is true.



In geometric terms it says that if any horizontal line $y = N$ is given between $y = f(a)$ and $y = f(b)$, then the graph of f cannot jump over the line. It must intersect $y = N$ somewhere.

Intermediate Value Theorem

Example 9

Show that there is a root of the equation

$$x^{201} - 5x^{200} + 6x^2 - 21x - 46 = 0$$

between 5 and 6.

Solution.

Let $f(x) = x^{201} - 5x^{200} + 6x^2 - 21x - 46$. We are looking for a solution of the given equation, that is, a number c between 5 and 6 such that $f(c) = 0$.

Therefore we take $a = 5$, $b = 6$, and $N = 0$ in the Intermediate Value Theorem.

We have

closed interval $[5, 6]$

$$f(5) = 5^{201} - 5 \times 5^{200} + 6 \times 5^2 - 21 \times 5 - 46 = -1 < 0$$

$f(5) < 0 < f(6)$

and

$$f(6) = 6^{201} - 5 \times 6^{200} + 6 \times 6^2 - 21 \times 6 - 46 = 6^{200} - 44 > 0$$

Example 9–Solution (Cont'd)

Thus $f(5) < 0 < f(6)$; that is, $N = 0$ is a number between $f(5)$ and $f(6)$.

Now f is continuous since it is a polynomial, so the Intermediate Value Theorem says there is a number c between 5 and 6 such that $f(c) = 0$.

In other words, the equation $x^{201} - 5x^{200} + 6x^2 - 21x - 46 = 0$ has at least one root c in the interval $(5, 6)$.

Comment: We can locate a root more precisely by using the Intermediate Value Theorem again. For instance, it is easy to check that $f(5.1) > 0$. Therefore, a root must lie between 5 and 5.1.

Continuity

Exercise

If $f(x) = x^2 + 10 \sin x$, show that there is a number c such that $f(c) = 1000$. **don't need to find exact value of c*

$$\left. \begin{array}{l} f(0) = 0 \\ f(50) > 1000 \end{array} \right\} \text{we can restrict the intervals more}$$

$$f(20) = 400 + 10 \sin 20 \leq 410$$

$-1 < \sin 20 < 1$

$$f(32) = 1024 + 10 \sin 32 \geq 1014$$

$\therefore$ By IVT, there is a number c between 20 and 32 such that $f(c) = 1000$.

Continuity

Prove $\lim_{x \rightarrow 4} \sqrt{x} = 2$

$\forall \varepsilon > 0$, find a $\delta > 0$, s.t.

if $-\delta < x - 4 < 0$, then

$$|\sqrt{x} - 2| < \varepsilon$$

Since $4 - \delta < x < 4$, $-\varepsilon < \sqrt{x} - 2 < \varepsilon$

$$|\sqrt{x} - 2| < \varepsilon \iff 2 - \sqrt{x} < \varepsilon$$

$$2 - \varepsilon < \sqrt{x} < 2 + \varepsilon$$

$$\text{or } \sqrt{x} > 2 - \varepsilon$$

① If $\varepsilon > 2$, any $\delta > 0$ works

they are both non-negative, so need to flip signs.

② If $\varepsilon \leq 2$, then $\sqrt{x} > 2 - \varepsilon \Rightarrow x > (2 - \varepsilon)^2 = 4 - (4\varepsilon - \varepsilon^2)$

$$\delta = 4\varepsilon - \varepsilon^2 = 4 - (2 - \varepsilon)^2$$

$$\delta = \begin{cases} 4 - (2 - \varepsilon)^2, & 0 < \varepsilon \leq 2 \\ 1, & \varepsilon > 2 \end{cases}$$

$\rightarrow$ chosen for simplicity

* similar to handout & slide 21